

ARRAYS 64

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Language Python 3



Problem

Given two sorted arrays, merge them and find the median of the combined sorted array.

Input Format

First line contains two integers n and m . Second line contains n sorted integers. Third line contains m sorted integers.

Constraints

$1 \leq n, m \leq 10^5$ $-10^6 \leq \text{array elements} \leq 10^6$

Output Format

Print the median of the merged array.

If the total number of elements is odd, print the middle element. If even, print the average of the two middle elements with one decimal place.

Sample Input 0

```
2 2
1 2
3 4
```

Sample Output 0

```
2.5
```

Sample Input 1

```
2 3
1 2
3 4 5
```

Sample Output 1

```
3.0
```

Sample Input 2

```
4 2
1 3 5 7
2 4
```

Sample Output 2

```
1 n, m = map(int, input().split())
2 arr1 = list(map(int, input().split()))
3 arr2 = list(map(int, input().split()))
4 i = 0
5 j = 0
6 merged = []
7 while i < n and j < m:
8     if arr1[i] <= arr2[j]:
9         merged.append(arr1[i])
10        i += 1
11    else:
12        merged.append(arr2[j])
13        j += 1
14 while i < n:
15     merged.append(arr1[i])
16     i += 1
17 while j < m:
18     merged.append(arr2[j])
19     j += 1
20 total = n + m
21 if total % 2 == 1:
22     print(float(merged[total // 2]))
23 else:
```

Line: 20 Col: 14



Upload Code as File

Run Code

Submit Code



Test against custom input

Congratulations

You solved this challenge. Would you like to challenge your friends?



Test case 0



Test case 1



Test case 2

Compiler Message

Success